STA 35C: Statistical Data Science III

Lecture 12: The Bias-variance Tradeoff

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Spring 2026, UC Davis

Announcements

Midterm 1 scores are posted on Canvas

- Exam and solutions are posted on the course webpage
- Please review the solutions carefully and identify what to revisit
- You can review graded exams in discussion section tomorrow

Post-midterm review

- Homework 3 will include selected problems from Midterm 1 with slight modifications
- The TA will go over Midterm 1 problems in discussion section tomorrow

Mid-course survey

- Please take 10 minutes to complete the survey on Canvas by Friday, May 1
- All feedback and constructive suggestions/requests are welcome

Agenda

Brief recap of pre-Midterm 1 topics

Today: Bias-variance tradeoff

- Model accuracy: MSE and classification error
- Training error vs. test error
- Bias, variance, and irreducible error
- Why overly simple and overly flexible models might both fail

Roadmap ahead: estimating test error, controlling model flexibility, and multiple testing

Recap: Supervised learning

$$\underbrace{X}_{\text{predictors}} \longrightarrow \underbrace{Y}_{\text{response}}$$

Goal: learn a rule or function $\hat{f}$ that uses X to explain or predict Y

- **Prediction:** use $\hat{f}(x_{\text{new}})$ to predict a future response
- **Inference:** understand how Y is related to the predictors

Two major problem types

- **Regression:** Y is quantitative
- **Classification:** Y is categorical

Key point: We fit models on training data, but want good performance on new test data

Recap: Probability and random variables

Probability language

- Outcome ω, sample space Ω, event $A \subseteq \Omega$
- Conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

- Bayes' theorem updates probabilities after observing evidence (posterior)

Random variables

- A random variable is a function $X : \Omega \to \mathbb{R}$
- PMF/PDF/CDF describe the distribution of X

Expectation and variance summarize center and spread

$$\mathbb{E}[X] = \sum_x x\, p_X(x) \qquad \to \qquad \mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] \qquad \to \qquad \mathrm{Var}(aX + bY + c) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\mathrm{Cov}(X, Y)$$

Recap: Regression

Regression setup

$$\underbrace{X}_{\text{predictors}} \longrightarrow \underbrace{Y}_{\text{quantitative response}}$$

Linear regression model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$$

- Coefficients are estimated by least squares

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- β_j is interpreted as the average change in the mean response per unit increase in X_j, holding other predictors fixed
- Standard errors, confidence intervals, and t-tests quantify uncertainty
- R^2, adjusted R^2, and RSE summarize model fit

Recap: Classification

Classification setup

$$\underbrace{X}_{\text{predictors}} \longrightarrow \underbrace{Y}_{\text{class label; qualitative}}$$

Logistic regression (discriminative)

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad p(x) = P(Y=1 \mid X=x)$$

- Predict class 1 if $\hat{p}(x) \geq p^*$
- Changing p^* changes the false positive / false negative tradeoff
- Confusion matrices, ROC curves, and AUC assess classification performance

Linear discriminant analysis (generative)

$$P(Y=k \mid X=x) = \frac{\pi_k f_k(x)}{\sum_j \pi_j f_j(x)}$$

- LDA assumes Gaussian class-conditional distributions with common covariance

Model accuracy: Regression vs. classification

Regression: Mean squared error (MSE)

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$$

- Smaller MSE means predictions are closer to observed responses
- Least squares minimizes training MSE, equivalently RSS

Classification: Error rate (=rate of misclassification)

$$\text{Error rate} = \frac{1}{n}\sum_{i=1}^{n} I(\hat{y}_i \neq y_i)$$

- Smaller error rate means fewer misclassified observations
- Other metrics may matter when false positives and false negatives have different costs

Training error vs. test error

Training error is computed on the same data used to fit the model:

$$\mathrm{MSE}_{\mathrm{train}} = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{f}(x_i) \right)^2$$

Test error measures performance on new, unseen data:

$$\mathrm{MSE}_{\mathrm{test}} = \frac{1}{n_{\mathrm{test}}} \sum_{j=1}^{n_{\mathrm{test}}} \left(y_j^{\mathrm{test}} - \hat{f}(x_j^{\mathrm{test}}) \right)^2$$

Key distinction

- Training error measures how well the model fits the data it saw
- Test error measures how well the model generalizes to new data
- We care most about test error, but it is usually unknown when choosing a model

The challenge in practice

Reality: We usually do not know test error when choosing a model before deployment

- We fit models using training data
- But the model with the smallest training error may not have the smallest test error

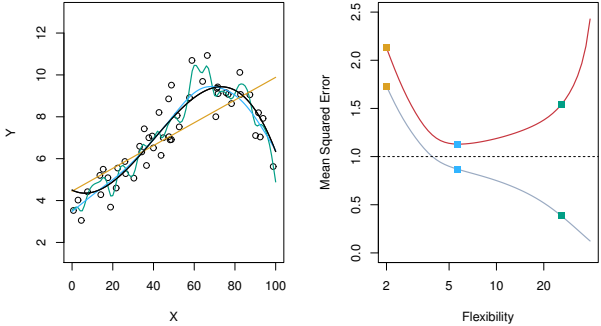

Figure: As model flexibility grows, training error usually decreases, while test error can eventually increase; Linear, Degree-5, Degree-25 regression [JWHT21, Figure 2.9].

Bias-variance tradeoff

At a fixed test point x_0, suppose a new response satisfies

$$Y_0 = f(x_0) + \varepsilon_0, \qquad \mathbb{E}[\varepsilon_0] = 0, \qquad \mathrm{Var}(\varepsilon_0) = \sigma^2.$$

The expected test error decomposes as

$$\mathbb{E}\left[\left(Y_0 - \hat{f}(x_0)\right)^2\right] = \underbrace{\left(f(x_0) - \mathbb{E}[\hat{f}(x_0)]\right)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}\left[\left(\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)]\right)^2\right]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible error}}$$

where the expectation is over both the training sample used to fit $\hat{f}$ and the new noise ε.

Interpretation

- **Bias:** systematic error from using a model class that is too simple
- **Variance:** sensitivity of $\hat{f}$ to the particular training sample
- **Irreducible error:** noise that no method can eliminate

Interpreting bias and variance

High bias

- Model is too simple to capture the true relationship
- Example: fitting a line when the true relationship is strongly curved
- Leads to underfitting: the model misses real structure in the X-Y relationship

High variance

- Model is too sensitive to small changes in the training data
- Example: fitting an overly flexible curve through noisy observations
- Leads to overfitting: the model fits noise rather than signal

Goal: Find the sweet spot

- Choose enough flexibility to reduce bias
- But not so much flexibility that variance becomes too large

Model flexibility and test error

As model flexibility increases:

- Training error usually decreases
- Test error may decrease at first, then increase
  - Bias tends to decrease
  - Variance tends to increase

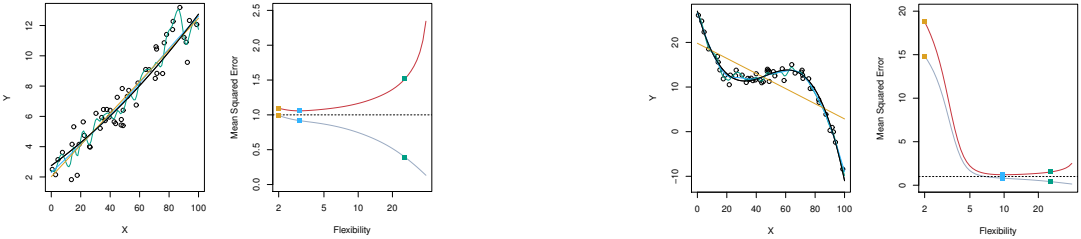

Figure: The same bias-variance idea appears across many settings [JWHT21, Figures 2.10 & 2.11].

$$\text{Expected test error} \approx \text{Bias}^2 + \text{Variance} + \text{Irreducible error}$$

Pop-up quiz: Training vs. test error

Three regression models are fit to the same training data and evaluated on a separate test set:

Model	Flexibility	Training MSE	Test MSE
A	*Low*	30	32
B	*Moderate*	18	20
C	*High*	6	28

Question: Which statement is most appropriate?

A) Choose Model C because it has the smallest training MSE.

B) Model A has the highest variance because it is the simplest model.

C) Test MSE always decreases as flexibility increases.

D) Choose Model B because it has the smallest test MSE.

Answer: D. Model C fits the training data best but likely overfits; Model B gives the best performance on new data among the three.

Summary of the bias-variance tradeoff

Overly simple models might underfit the data

- High bias
- Low variance

Overly flexible models might overfit the training data

- Low bias
- High variance

Best predictive model often lies between these extremes

- Balances bias and variance to minimize test error
- Should be chosen using evidence about generalization, not training error alone

Looking ahead: Roadmap until Midterm 2

Main theme: Building models that fit signal, not noise, thereby generalizing well

The bias-variance tradeoff raises several practical questions.

1. **How can we estimate test performance using training data only?**
   - Cross-validation
2. **How can we quantify uncertainty when formulas are difficult?**
   - Bootstrap
3. **How can we allow flexibility but avoid overfitting?**
   - Model selection and regularization
4. **When many predictors or tests are considered, which findings are reliable?**
   - Multiple hypotheses testing and false discovery control

Unifying question: How do we learn useful patterns without mistaking noise for signal?

Wrap-up

- Before Midterm 1, we focused on probability, regression, classification, and interpreting fitted models.
- After Midterm 1, the central question becomes: *How well will a model perform on new data?*
- Training error measures fit to observed data; test error measures performance on new data.
- More flexibility can reduce bias, but too much flexibility can increase variance and cause overfitting.
- The bias-variance tradeoff explains why the model with the smallest training error is not necessarily the best predictive model.
- Upcoming lectures cover tools—cross-validation, bootstrap, regularization, and multiple-testing adjustments—that help us estimate performance, control overfitting, and quantify uncertainty.

References

Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.
An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.
Springer, New York, NY, 2nd edition, 2021.